Six clues from powder to path

How a Sidewalk Is Made and How It Works

Name _____ Date _____



Big and Little found a chipped corner on a sidewalk square. Inside, tiny stones and sandy specks sat packed together. Beside the chip stood four sealed jars: gravel, sand, a gray powder called cement, and water. A sidewalk is made of concrete, and concrete is a mix of all four.



Little held a magnifier up to an old chunk of concrete. The stones sat inside smooth gray paste, like raisins in a muffin. Cement mixed with water makes that paste. The paste coats every grain of sand and every stone, and then it holds them tight.



Next came the pour. Wet concrete filled a wooden frame called a form, and then it was spread flat and smoothed. Big and Little watched from behind a rope, because wet concrete is not for touching. Soon the cement and water began a chemical reaction, a change that links them into something solid. Concrete hardens by that reaction, not just by drying, so it can even harden under water.

concrete	a hard mix of cement, water, sand and gravel
cement	a fine gray powder that makes a paste when it meets water
form	a wooden frame that holds wet concrete in shape
chemical reaction	a change that makes something new, like cement and water linking together
curing	keeping new concrete damp while it gets stronger
groove	a thin line pressed into a slab to guide where a crack will go

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A new slab is not finished when it looks hard. It stays damp for about a week so the reaction can keep going, and most of its strength builds up in the first month. Big lined up seven pebbles on the frame, one for each day under the damp cloth. Keeping new concrete damp like this is called curing.



Concrete shrinks a little as it hardens, and heat and cold make it grow and shrink all year. Because concrete does not like bending or stretching, it can crack. So workers press a groove across each square. The groove makes the slab thinner along one line, so a crack tends to start there, neat and straight, instead of anywhere at all.



At the edge of a park path, Little spotted a thin layer of packed gravel under the slab. Concrete is very strong when it is squeezed, and feet and bike wheels squeeze it all day. It is weaker when it bends. The gravel helps support the slab evenly, so the weight spreads out and the slab bends less.

1. Why is cement only one ingredient of concrete, and what does it do when it meets water?
2. How is hardening by a chemical reaction different from drying, and why does curing matter?
3. Why do sidewalks have grooves, and how does a gravel base help a slab?

Your turn

Write one sentence that follows concrete from powder to a hard slab.

Example: Cement and water make a paste that holds the stones, so the slab turns hard.